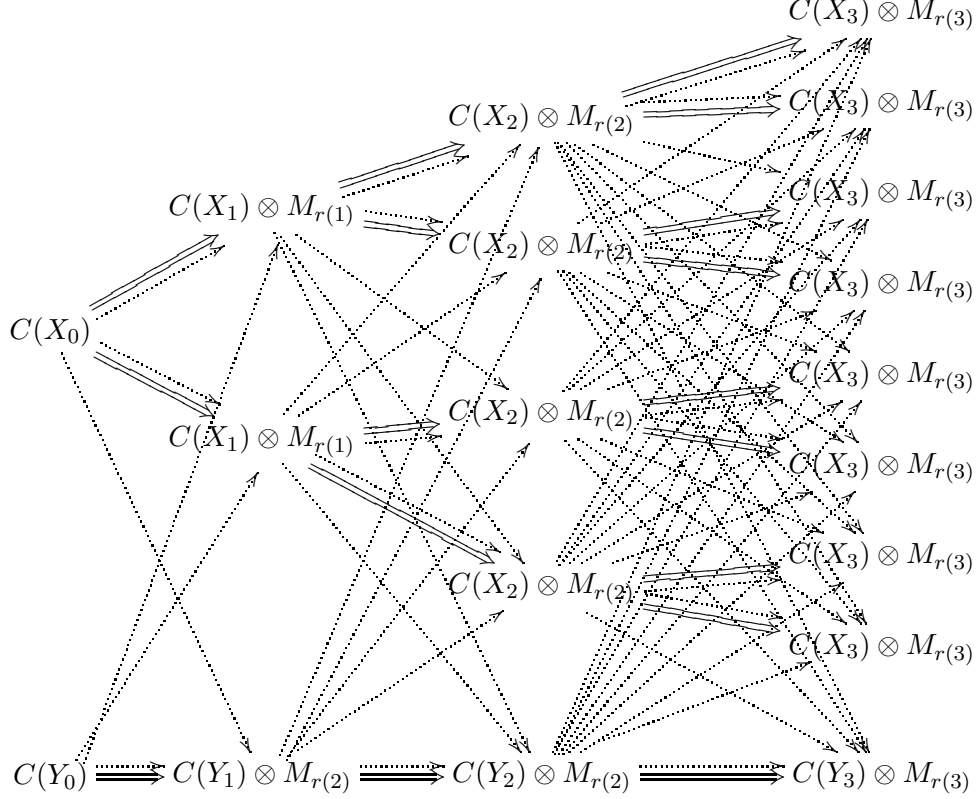


(1.2)



The radius of comparison of the resulting AH algebra is shown to be the maximum of the two. Up to composing with inner automorphisms, which are needed to ensure consistency, the actions on the top levels are cyclic permutations, whereas the action on the bottom part is trivial. In essence, we are merging an action which is a souped-up version of the odometer action - which has the Rokhlin property - with an inner action on the bottom part. Since no general result applies here, we analyze the crossed product directly: the crossed product is an AH algebra too, and its radius of comparison is shown to be the same as the one coming from the bottom row.

The rest of the paper constitutes the proof of Theorem 1.1. In light of Remark 1.2, we only prove explicitly the case in which $0 < r' \leq r \leq \infty$. We divide the proof into sections. In Section 2, we recall some definitions and prove an elementary lemma about the existence of sequences of natural numbers with a prescribed growth condition. In Section 3 we construct our AH algebra A , along with the action of $\mathbb{Z}^d$. In Section 4 we compute the radius of comparison of A , and in Section 5 we compute the radius of comparison of the crossed product.